Diego Armando Salinas Lugo

AI & Machine Learning Engineer · Founder

Monterrey, Mexico · TN visa eligible · English / Spanish

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PROFILE

AI & ML engineer who builds full systems, not notebooks. Generative AI at SAP for automated technical fault reporting, and founder of ARMATUS — a shipped iOS coach that architects training weeks from profile, readiness and logged evidence. MSc Artificial Intelligence with Distinction. Work spans signal processing, explainable clinical ML, predictive maintenance, RAG and deployed FastAPI / Streamlit / Cloud Run products.

EXPERIENCE

Machine Learning & AI Engineer · SAP
Jan 2026 — Present

San Pedro Garza García, NL

- Designs and deploys Generative AI that automates technical fault reporting — improving incident categorization and cutting response time.
- Builds on SAP Business AI and BTP extension patterns, consuming leading LLMs through SAP AI Core, AI Launchpad and the generative AI hub.
- SAP Certified — SAP Generative AI Developer (2026).

Founder & Developer · ARMATUS · armatus.app
May 2026 — Present

- Founded and shipped a consumer generative-AI iOS training coach, live at armatus.app.
- Deep onboarding — goals, equipment, injuries, training history, lifestyle — builds the athlete model that drives every plan.
- Daily signals (energy, sleep, available time) plus logged sessions feed an overload / readiness loop that regenerates each week from evidence.
- Readiness ceiling is checked before load; hybrid sport (running, cycling, swimming) is counted as real fatigue, not a footnote.
- Production iOS client and streaming API for structured plan generation, bilingual ES / EN.

Data & AI Trainee · Interius
Sep 2025 — Dec 2025

San Pedro Garza García, NL

- Delivered AI MVPs end to end: strategy, analysis, feature engineering, model training, evaluation and deployment.
- Built advanced contact segmentation and workload / efficiency diagnostics used for decision-making.
- Shipped as containerized Streamlit services on Google Cloud Run.

AI Student Researcher · University of Essex
Nov 2024 — Sep 2025

Colchester, UK

- End-to-end ML for emotion recognition from HRV features derived from PPG across 62 participants.
- Butterworth filtering, IBI extraction, and time / frequency / non-linear (Poincaré) feature engineering.
- Leave-one-participant-out validation and model comparison; Random Forest reached r = 0.5975 for continuous arousal.

SELECTED SYSTEMS

Engines Health Monitor — Predictive Maintenance
XGBoost RUL on NASA C-MAPSS with SHAP attribution, FastAPI inference, Streamlit fleet dashboard, and a LangGraph multi-agent layer for NL diagnostics and PDF briefings. Deployed on Google Cloud Run.

Test RMSE 16.74 cycles · MAE 12.33 · R^2 0.825 rul-dashboard-368785016309.us-central1.run.app

Personal Research Assistant (RAG)
PDF question answering with chunking, embeddings and a MultiQueryRetriever; GPT-4o answers grounded in ChromaDB with visible source snippets. FastAPI + Streamlit.

149 indexed segments · every answer cites its source github.com/DArmandoSalinas/RAG

Cardiovascular Risk Classification
Leak-free clinical pipeline on UCI four-site heart data (920 patients); sentinel-zero repair, ColumnTransformer inside CV, HistGradientBoosting, SHAP + LIME, Leave-One-Site-Out transportability check.

ROC-AUC 0.90 · recall 87.3% · F1 0.85 · LOSO mean AUC 0.79 github.com/DArmandoSalinas/HeartDisease-Predictor

ADHD & Sex Prediction with Fairness
1,213 participants, fMRI connectomes and psychosocial data; MI selection, KernelPCA, Random Forest and Logistic Regression with LIME / SHAP and fairness reporting by sex.

ADHD accuracy 82% · recall 0.88–0.90 · AUC 0.866 github.com/DArmandoSalinas/Predicting-ADHD-sex

WellHave — Burnout prediction + LLM coach
Burnout classifier (low / moderate / high) paired with GPT-4o coaching via LangChain. FastAPI backend, React Native (Expo) client, Supabase.

github.com/DArmandoSalinas/Wellhave

Motor Performance Monitoring
Serial ingestion from a Sumitomo three-phase motor and Haas Mini Mill; machine-specific baselines, adaptive z-score and temperature-slope thresholds, 0–100 health scores in Streamlit. Fully interpretable rules.

github.com/DArmandoSalinas/Motor-performance-prediction

Text Emotion Recognition
XGBoost over ~416k tweets across six emotion classes with bag-of-words, stemming and stratified K-fold.

87.5% accuracy github.com/DArmandoSalinas/Leveraging-XGBoost-for-Text-Classification-on-Emotion-Recognition

Hybrid Movie Recommender
SVD collaborative filtering blended with TF-IDF content similarity behind a Flask interface.

SVD RMSE 0.477 vs 3.624 baseline github.com/DArmandoSalinas/movie-recommendation-system

Politrauma

Interactive clinical-education web tool for polytrauma assessment workflows. Educational, not a medical device.

politrauma.vercel.app

SKILLS

ML	Python · scikit-learn · XGBoost · TensorFlow / Keras · SHAP · LIME · NeuroKit2 · SciPy · MLflow
PLATFORM	FastAPI · LangChain · LangGraph · OpenAI · SAP AI Core / generative AI hub · Streamlit · Flask · Docker · GCP Cloud Run · Vercel · TypeScript · React Native / Expo · SQL · Git
DOMAIN	Signal processing (PPG, HRV, vibration) · Predictive maintenance · Healthcare ML & fairness · Retrieval-augmented generation · Robotics · PID · fuzzy control · MATLAB · C

EDUCATION

MSc Artificial Intelligence — with Distinction
2024 — 2025

University of Essex, UK

Thesis: A Machine Learning Approach for Emotion Recognition Using Heart Rate Variability.

B.S. Mechatronics Engineering
2021 — 2025

Tecnológico de Monterrey, MX

Control systems, industrial automation, robot design, manufacturing.

CERTIFICATIONS

- SAP Certified — SAP Generative AI Developer · SAP, 2026
- Google Analytics · Google Digital Academy, 2025
- Introduction to Generative AI · Google Cloud, 2024
- Intro to Machine Learning · Kaggle, 2024
- CS50's Introduction to Computer Science · Harvard University, 2022
- Databases and SQL for Data Science with Python · IBM, 2026
- Machine Learning Specialization · Stanford Online & DeepLearning.AI, 2024
- AI For Everyone · DeepLearning.AI, 2024
- 2023 NanoLab (online) · MIT / Tec-MIT Nanotechnology, 2023
- MATLAB Onramp · MathWorks, 2022